



A novel framework for critical heat flux prediction based on multiphase CFD-informed neural networks

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ABSTRACT

The Critical Heat Flux (CHF) represents a fundamental thermohydraulic parameter for ensuring the safe operation of thermal systems involving boiling. Recent advancements in multiphase numerical models have established Computational Fluid Dynamics (CFD) as an essential instrument for analyzing boiling phenomena. Nonetheless, the absence of a universal consensus regarding the triggering mechanisms of CHF has resulted in diverse CHF criteria, causing considerable uncertainties in CHF predictions. This study employs a previously developed heat flux partitioning model that directly represents the physical principles of boiling heat transfer across temporal and spatial dimensions. It predicts the CHF for flow boiling in a rectangular channel. A comparative analysis of various CHF criteria on prediction accuracy was conducted. Furthermore, this study applied a neural network framework enhanced with prior domain knowledge derived from the multiphase CFD information. Bayesian optimization was utilized to identify the optimal hyperparameters. A thorough comparative analysis was conducted among the proposed network framework, traditional standalone neural networks, and multiphase CFD methods. This analysis leveraged 433 data points from the 2006 CHF lookup table, encompassing various flow conditions: pressures between 7 and 18 MPa, mass fluxes ranging from 750 to 5000 kg/(m²·s), and qualities varying from -0.2 to 0.25. The results indicate that the multiphase CFD-informed neural network framework exhibits remarkable generalization abilities across the test dataset, particularly regarding extrapolation. Additionally, a comprehensive comparison is presented between the novel framework and the physics-informed machine learning approach, which incorporates empirical correlations as prior knowledge. The proposed multiphase CFD-informed neural network aims to support future experimental efforts and facilitate the analysis of thermodynamic systems pertinent to the boiling crisis.

1. Introduction

The CHF (Critical Heat Flux) crisis is a common issue in various applications involving thermodynamic systems, including vapor chambers, thermal management for batteries, cooling of high-power-density electronic devices and nuclear power plants. These applications typically enhance heat transfer by relying on bubble growth and departure from nucleation sites periodically, a phenomenon known as nucleate boiling. The complex bubble dynamic behaviors cause the heat transfer mechanism at the boiling surface to depart from nucleate boiling when CHF occurs. This disruption causes a rapid decrease in the local heat transfer coefficient, leading to a sharp increase in wall temperature. Consequently, there is a potential for the failure of the thermodynamic

system, which significantly raises the risk of severe safety incidents.

The significance of CHF has prompted extensive in-depth experimental and theoretical research efforts over the past few decades. This has resulted in the development of various mechanistic models for CHF, including concepts such as bubble crowding [1], liquid sublayer dry-out [2], and, more recently, models based on percolation theory [3,4], as well as thermo-fluidic and stochastic interactions [5]. Despite advancements, existing mechanistic models still have gaps in understanding the triggering mechanisms of CHF. These gaps arise from limitations in measurement techniques at the characteristic scale of CHF in experimental methods [6] and from inadequacies in the representation of bubble dynamics in mathematical models [7,8]. Therefore, numerous data-driven empirical correlations continue to play a crucial

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role in predicting CHF [9,10]. However, conducting CHF-related experiments is both resource-intensive and time-consuming.

Engineers are continually exploring innovative and reliable approaches for predicting CHF. Recent progress in validated/verified phase change models has enabled the application of CFD (Computational Fluid Dynamics) techniques to explore the intricate local characteristics of essential physical parameters and boiling crisis within multiphase and multiscale thermodynamic systems. CFD techniques demonstrate enhanced cost-effectiveness and provide more granular and localized parameters for analyzing two-phase systems compared to lumped parameter methods. Leveraging the CFD method for CHF prediction during the early stages of the design of the thermodynamic system remains a fundamental objective.

Sensitivity analyses performed on the parameters within the bubble dynamic model [11,12] and the development of advanced wall boiling models [13–15] have established a strong basis for predicting CHF based on multiphase CFD techniques. Extensive research has been conducted regarding the boiling crisis that manifests within various geometric configurations. Wang [16] and Tayerian [17] investigated the effects of non-uniform heat flux input on CHF in upward pipes, employing the RPI model [18] as a foundational framework for the analysis. Additionally, Rao [19] focus on the influence of eccentricity on CHF within annular channels. Liu et al. [20,21] explored the impact of spacer grids and mixing vanes on CHF for both single-rod and 5×5 rod bundle configurations. Zhang [22] and Brewster [23] performed CFD analyses to predict CHF in simplified fuel assemblies and typical CANDU fuel bundles, focusing specifically on the operational conditions typical of pressurized heavy water reactors.

Additionally, researchers have developed advanced heat flux partitioning models that incorporate recent physical insights to enhance the prediction of CHF. Zhang [24,25] employed the Demarly boiling model [14], which characterizes the boiling crisis through the irreversible dry-patch evolution process, to predict both the magnitude and location of CHF in a 5×5 fuel assembly configuration. Cong [26] enhanced the existing RPI model by incorporating a thin-film heat flux term for similar geometries, which improved the accuracy of CHF predictions.

Despite numerous studies employing multiphase CFD methods for CHF prediction with reasonable accuracy, significant debates and controversy persist regarding the criteria to determine CHF. Table 1 presents a brief overview of the various methods for diagnosing CHF reported in the related literature. Researchers typically employ methods similar to those found in CHF experiments. They track the maximum wall temperature during CFD simulations to identify the heat flux at which a significant increase occurs, marking the CHF point. For instance, in the two-phase CFD studies conducted by Taherian [17], Zhang [24,25], Vadlamudi [27–29], and Dione [30], the rapid increase in wall temperature serves as the criterion for determining CHF. Additionally, certain researchers propose that the CHF point can be identified as the initial point where the boiling curve deviates from linearity. This perspective aims to improve the convergence of CFD calculations and is supported by studies conducted by Liu [20,21], Cai [39], Liu [40] and Li [41].

The data-rich characteristics of CFD methods provide a more effective means of acquiring localized two-phase physical quantities compared to traditional experimental approaches. This attribute significantly enhances the development of CHF prediction methods relying on local parameters. Numerous researchers have proposed that CHF is triggered when the local void fraction at the boiling surface exceeds a specific critical threshold, rooted in the bubble crowding theory originally presented by Weisman and Pei [1]. Despite this theoretical foundation, there is still no consensus within the community on the exact quantification of this critical threshold value.

The traditional bubble crowding theory defines the CHF as the moment when ellipsoidal bubbles, characterized by an aspect ratio of 3:1, completely occupy the bubble layer. This state is associated with a local void fraction of 0.82, which some researchers advocate as a key

Table 1

Overview of the criteria for determining CHF.

Author(s)	Geometry Configuration	CHF Criteria	Remarks
Taherian [17]	Pipe	Rapid increase in wall temperature	Liner function
Vadlamudi [27–29]	Pipe		Liner function
Dione [30]	Pipe		Liner function
Zhang [31]	2×2 rod bundle		Liner function
Zhang [32]	5×5 rod bundle	Deviation of boiling curve from linearity	Liner function
Pothukuchi [33]	Eccentric annulus and 4×4 rod bundle		Liner function
Liang [34]	IVR-ERVC		Quadratic function
Du [35]	5×5 rod bundle		Cubic function
Su [36]	5×5 rod bundle		Cubic function
Podila [37]	Pipe with buttons		Cubic function
Zhang [24]	5×5 rod bundle		Dry-patch model
Zhang [25]	Hexagonal shape fuel assembly		Dry-patch model
Cong [26]	Pipe and 5×5 rod bundle		Info not supplied
Li [38]	Narrow rectangular channel		Info not supplied
Cai [39]	3×3 petal-shaped rod bundle	Attainment of critical void fraction threshold	Liner function
Liu [20]	Pipe and single-rod bundle		Cubic function
Liu [21]	5×5 rod bundle		Cubic function
Liu [40]	5×5 rod bundle		Cubic function
Li [41]	Pipe	Evaporation heat flux proportion attaining a critical value	Cubic function
Vyskocil [42]	Pipe		0.80
Zhang [22]	Hexagonal shape fuel assembly		0.82
Kim [43]	Pipe		0.90
Zhang [44]	5×5 rod bundle	CHF indicator $(\frac{T_{w\alpha}}{T_s})$ achieving a value of 1	0.95
Brewster [23]	CANDU fuel bundles		0.99
Vyskocil [46]	Pipe	CHF indicator $(\frac{T_{w\alpha}}{T_s})$ achieving a value of 1	–
Xu [47]	5×5 rod bundle		–

criterion for determining CHF in multiphase CFD simulations. For instance, Zhang [22] applied this threshold to CHF prediction for fuel assemblies, achieving a prediction accuracy with a deviation of less than $\pm 20\%$. Under the hypothesis that bubbles can be modelled as non-deformable spheres, the close-packing theory indicates that the maximum critical void fraction should not exceed 0.74. Meanwhile, the maximum 2D packing theory permits a higher critical void fraction, with a theoretical upper limit of 0.9069. In practice, the critical void fraction threshold in current CFD-based methods ranges from 0.8 to 0.9 [42,43]. Notably, Zhang [44] and Brewster [23] propose even higher thresholds, suggesting critical void fractions of 0.95 and 0.99, respectively.

The definition of bubble layer thickness remains a topic of debate among researchers. Generally, the thickness is defined by the wall-adjacent cells [35,36]; however, this approach can exhibit considerable sensitivity to the wall-adjacent cells [12]. Certain studies adopt a fixed y^+ to determine the thickness of the bubble layer to reduce the impact of wall-adjacent on CHF criteria [39,45]. Brewster [23] improved the accuracy of CHF predictions by implementing a calibration process for bubble layer thickness. Additionally, some researchers

propose that a critical threshold of the proportion of evaporation heat flux is a key mechanism for triggering CHF [46]. Research conducted by Westinghouse Electric Company [47] indicates that CHF occurs when the CHF indicator reaches a value of 1.

Given the current understanding of CHF, addressing the issue of repetitive calibration in CFD-based CHF prediction methods from the first principles is impractical. In this light, neural networks could serve as a more dependable tool for managing the complex and chaotic nature of CHF.

Recent advancements in the application of neural network methodologies have significantly contributed to the understanding of boiling phenomena. Notably, Sinha [48] leveraged convolutional neural networks (CNNs) to predict acoustic emission characteristics associated with pool boiling. They further employed these sound patterns to predict CHF. In the research conducted by Rassoulinejad-Mousavi [49], an image-based deep-learning approach was used to predict CHF in pool boiling scenarios. Furthermore, Sajjad [50] integrated Bayesian optimization with deep neural networks to model the boiling curve leading up to the CHF stage. Additionally, in the context of flow boiling, the application of Physics-informed Machine Learning (PIML) has garnered attention. This approach works by learning patterns from the deviations between prior theoretical models and real-world data. The work by Zhao [51], Niu [52], and Qiu [53] exemplifies the effectiveness of this methodology for predicting CHF in flow boiling systems. These studies collectively emphasize the potential of advanced machine learning techniques in enhancing the predictive capabilities related to boiling crises.

In brief, the path to accurate prediction of CHF remains challenging. Globally, datasets related to CHF are scarce and sparingly distributed. Despite extensive research efforts spanning decades, detailed insights into local bubble dynamics associated with CHF remain sparse. The introduction of extraneous input variables can precipitate severe failures in data-driven methodologies, including conventional neural networks and lumped parameter models. On the other hand, CFD methods, which integrate significant prior domain knowledge, face controversies regarding the criteria for CHF. Current research lacks a quantitative analysis of these controversies.

To address these challenges, this study proposes a hybrid framework integrating multiphase CFD model with neural networks. As illustrated in Fig. 1, our methodology follows three key phases:

- 1) A multiphase CFD model [15] is developed to predict CHF based on physical boiling mechanisms, while systematically quantifying uncertainties arising from diverse CHF criteria (Section 2).
- 2) A deep feedforward neural network framework was developed to systematically learn the deviations between multiphase CFD predictions (informed by bubble dynamics models, e.g., wall heat flux partitioning and critical void fraction thresholds) and experimental CHF values. This framework implicitly encodes the influence of uncalibrated bubble dynamics parameters through residual learning, leveraging Bayesian optimization for hyperparameter tuning (Section 3).
- 3) The novel framework is rigorously evaluated against standalone neural networks and physics-informed machine learning, emphasizing its extrapolation capabilities and interpretability. (Section 4).
- 4) The primary objective is to establish an interpretable, physics-constrained framework for CHF prediction beyond purely data-driven or CFD methods.

2. Multiphase CFD framework

2.1. Eulerian-Eulerian two-fluid model

The Eulerian-Eulerian two-fluid model serves as a foundational framework to quantify the dynamics of liquid-vapor behavior. It conceptualizes each phase as an interpenetrating continuum with a shared pressure field. The model addresses conservation equations for mass, momentum, and energy for each phase independently. A comprehensive derivation of these equations can be found in our earlier publication [12, 15].

2.2. Fundamentals of physics-based boiling models

The heat flux partitioning model generates mass source terms from the liquid-vapor phase change that occurs due to wall boiling at wall-adjacent cells. This modelling approach bridges the gap between the mechanisms of boiling at the microscale with multiphase CFD models at the macroscale. The recently proposed heat flux partitioning model [15] provides a quantitative framework for analyzing the heat transfer phenomena associated with bubble dynamics during the boiling process, addressing both temporal and spatial dimensions. As illustrated in Fig. 2, the model defines the wall heat flux as the sum of five distinct

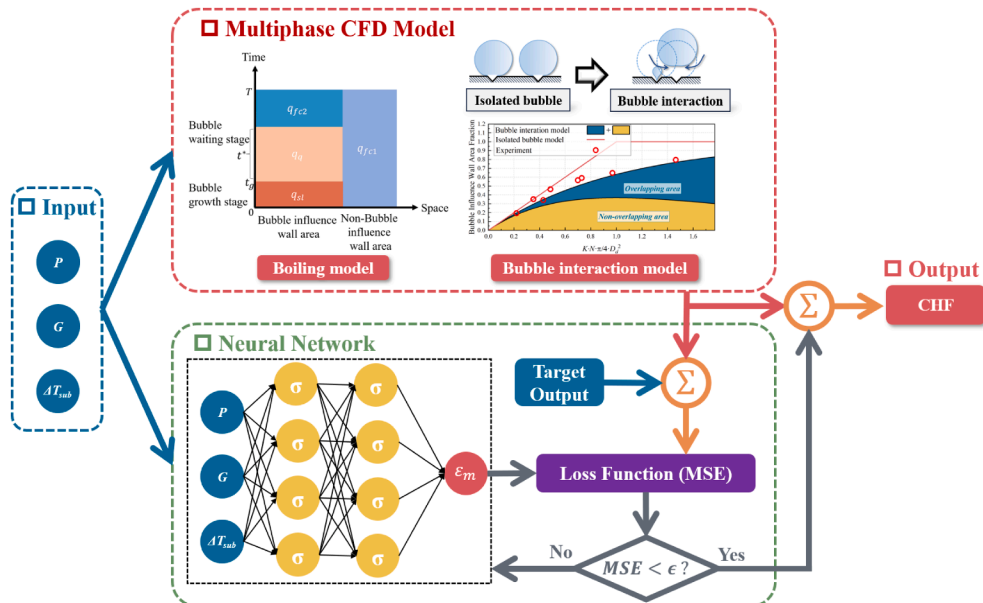


Fig. 1. Bridging multiphase CFD model and machine learning: A hybrid framework for CHF prediction.

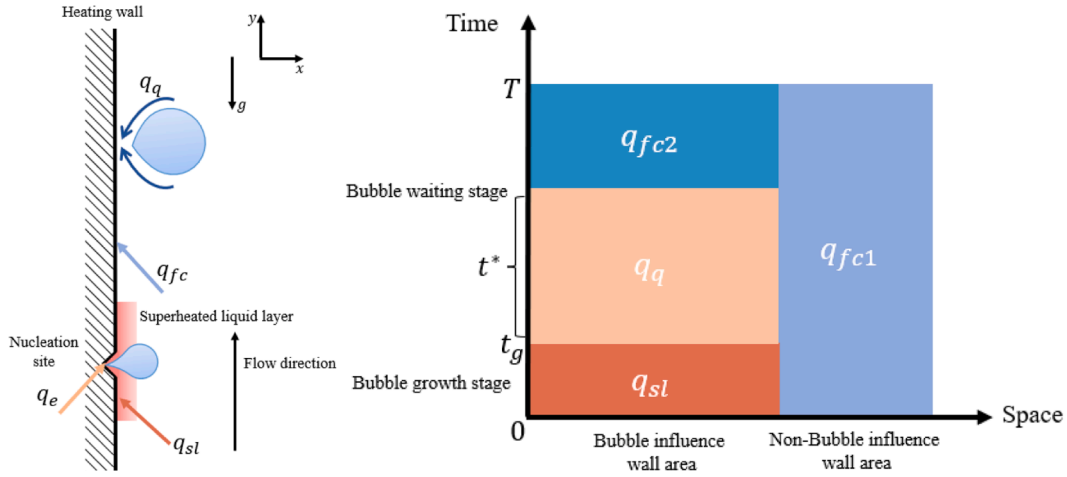


Fig. 2. Boiling heat transfer mechanism of the heat flux partitioning model [15].

components:

$$q_w = q_{fc1} + q_{fc2} + q_q + q_{sl} + q_e \quad (1)$$

The convection heat flux in the model has two components: q_{fc1} , which relates to the non-bubble influence area, and q_{fc2} , which pertains to the bubble waiting stage in the bubble influence area. The q_{fc1} is independent of bubble dynamics and represents a spatial proportion of $(1 - K_q)$ and a temporal proportion of 1, as formulated below:

$$q_{fc1} = (1 - K_q) h_{fc} (T_w - T_l) \quad (2)$$

where K_q and h_{fc} represent the bubble influence area and liquid convection heat transfer coefficient.

The term q_{fc2} guarantees that the minimum heat transfer coefficient adjacent to the boiling wall matches the liquid-phase convection coefficient. If the characteristic time (t^*) is longer than the bubble waiting time (t_g), then q_{fc2} is zero. Conversely, q_{fc2} receives a spatial allocation proportion of K_q and a temporal allocation of $(1 - t_w f - t^* f)$.

$$q_{fc2} = K_q \cdot \max[0, (1 - t_w f - t^* f)] h_{fc} (T_w - T_l) \quad (3)$$

where f denotes the bubble departure frequency, calculated as the inverse of the total time taken for bubble growth and departure.

The term q_q refers to the quenching heat flux, which is caused by the regeneration of the boundary layer initiated by the cool liquid from the bulk that occupies the space vacated by the departing bubble. The temporal proportion of this heat flux is determined by the bubble waiting time (t_w) or characteristic time (t^*), depending on the relative durations of these two factors. The spatial proportion corresponds to the bubble influence area, taking into account bubble interactions.

$$q_q = \begin{cases} \frac{1}{t_g + t_w} \int_0^{t^*} \frac{\lambda_l (T_w - T_l)}{\sqrt{\pi \eta_l t}} dt \cdot (K_{q, \text{Non-overlap}} + 2K_{q, \text{overlap}}) & t^* < t_w \\ \frac{1}{t_g + t_w} \int_0^{t_w} \frac{\lambda_l (T_w - T_l)}{\sqrt{\pi \eta_l t}} dt \cdot (K_{q, \text{Non-overlap}} + 2K_{q, \text{overlap}}) & t^* \geq t_w \end{cases} \quad (4)$$

where $K_{q, \text{Non-overlap}}$ and $K_{q, \text{overlap}}$ denote the non-overlapping domain and overlapping domain within the bubble influence area, respectively.

The term q_{sl} refers to the enhanced heat transfer near the boiling surface during the inertia-controlled stage of bubble growth, where rapid bubble expansion disrupts the superheated layer. This heat flux allocated a temporal proportion of bubble growth time and a spatial proportion concerning the bubble influence area while also considering interactions between the bubbles.

$$q_{sl} = \frac{1}{t_g + t_w} \int_0^{t_g} \frac{\lambda_l}{\sqrt{\pi \eta_l t}} (T_w - T_l) dt \cdot (K_{q, \text{Non-overlap}} + 2K_{q, \text{overlap}}) \quad (5)$$

where η_l and λ_l represent the liquid thermal diffusivity and conductivity, respectively.

Evaporation heat flux is driven by the latent heat necessary for nucleation bubble formation and is significantly affected by parameters related to bubble dynamics. Mathematically, it can be expressed as:

$$q_e = \frac{\pi}{6} \rho_g h_{lg} N f D_d^3 \quad (6)$$

where ρ_g , h_{lg} , N , and D_d refer to vapor density, latent heat, nucleation site density and bubble departure diameter, respectively.

2.3. Bubble dynamic parameters for interphase interaction

2.3.1. Boiling sub-models

The heat flux partitioning model requires accurate modelling of bubble dynamics parameters, such as nucleation site density, bubble departure diameter, and frequency. The selection of sub-model combinations significantly affects multiphase CFD simulation involving boiling. Ongoing challenges include developing universally applicable sub-model sets for various flow conditions [11,12]. This research employs sub-models grounded in data-driven empirical correlation derived from Massachusetts Institute of Technology (MIT) series experiments to reduce uncertainties, with nucleation site density directly derived from experimental data.

The research utilizes an empirical correlation established by Komajnosyula [55] to define the bubble departure diameter, mathematically expressed as follows:

$$D_d = 18.9 \times 10^{-6} \cdot \left(\frac{\rho_l - \rho_g}{\rho_g} \right)^{0.27} Ja_{\text{sup}}^{0.75} (1 + Ja_{\text{sub}})^{-0.3} \cdot u_l^{-0.26} \quad (7)$$

where ρ_l refers to the liquid density, while u_l signifies the liquid velocity. The terms Ja_{sup} and Ja_{sub} denote the superheated and subcooled Jacob numbers, respectively. These are quantitatively defined as:

$$Ja_{\text{sup}} = \frac{\rho_l c_{pl} (T_w - T_{\text{sat}})}{\rho_g h_{lg}} \quad (8)$$

$$Ja_{\text{sub}} = \frac{\rho_l c_{pl} (T_{\text{sat}} - T_l)}{\rho_g h_{lg}} \quad (9)$$

The bubble departure frequency is the inverse of the bubble

ebullition cycle, which encompasses the entire duration required for bubble nucleation, growth, departure, and subsequent quenching.

$$f = \frac{1}{t_g + t_w} \quad (10)$$

Bubble growth time t_g and bubble wait time t_w are represented as follows:

$$t_g = \frac{D_d^2}{4K^2} \quad (11)$$

$$t_w = \frac{0.0061 \times Ja_{\text{sub}}^{0.63}}{T_w - T_{\text{sat}}} \quad (12)$$

K represents the growth constant in the bubble growth model, which can be expressed mathematically as:

$$K = \frac{\sqrt{\eta_l}}{0.804\sqrt{\text{Pr}_l}} Ja_{\text{sup}} + \chi \cdot 1.95 Ja_{\text{sup}} \eta_l \quad (13)$$

where χ represents the condensation effect at the top of the bubble during its growth stage, formulated as follows:

$$\chi = 1.55 - 0.05 \frac{T_{\text{sat}} - T_l}{T_w - T_{\text{sat}}} \quad (14)$$

2.3.2. Bubble influence area

The "bubble influence area" is a factor within the heat flux partitioning model that characterizes the portion of the heated surface controlled by boiling. It identifies the area of bubble nucleation and quantifies the influence range of the nucleated bubbles. This parameter typically defined by isolated bubble model as follows:

$$K_q = K \frac{\pi}{4} D_d^2 \cdot N \quad (15)$$

The latest research [15] presents a framework for assessing the bubble influence area in scenarios marked by interaction among bubbles. Increased heat flux leads to a significant rise in nucleation site density. This increase contributes to a reduction in the bubble influence area attributed to the overlapping effects of adjacent bubbles, as illustrated in Fig. 3(a). This bubble interaction model is grounded in the inherent characteristics of the random distribution of nucleation sites on

the boiling surface. Fig. 3(b) illustrates that the influence area associated with each bubble is constrained to a range corresponding to one bubble departure diameter (i.e., $K = 1$). This value is derived from visualization experiments conducted by Amidu [56].

The bubble interaction model [15] enables the quantification of both overlapping and non-overlapping areas within the bubble influence area. Mathematically, this can be expressed as follows:

$$K_q = 1 - e^{-K \cdot N \frac{\pi}{4} D_d^2} \quad (16)$$

$$K_{q, \text{Non-overlap}} = e^{-K \cdot N \frac{\pi}{4} D_d^2} \cdot K \cdot N \cdot \frac{\pi}{4} D_d^2 \quad (17)$$

$$K_{q, \text{overlap}} = 1 - e^{-K \cdot N \frac{\pi}{4} D_d^2} - e^{-K \cdot N \frac{\pi}{4} D_d^2} \cdot K \cdot N \cdot \frac{\pi}{4} D_d^2 \quad (18)$$

Fig. 3(c) presents the validation of the proposed model [15]. It is evident that the isolated bubble model tends to overestimate the bubble influence area, primarily due to its assumption of negligible overlap between adjacent bubbles. In extreme scenarios, this model is restricted to a maximum value of 1 to maintain physical plausibility. CHF occurs in regions characterized by intense bubble interactions. Thus, it can be inferred that the bubble interaction model [15] is more adept at representing the influence area, particularly under CHF conditions.

2.4. Critical heat flux modeling

2.4.1. Wall dry-out area fraction

During the transition from liquid convection to the boiling crisis, the heat transfer mechanism at the heated wall progresses through a gradual evolution from nucleate boiling to vapor convection. The concept of "Wall dry-out area fraction" serves as a quantifiable parameter for characterizing this transition, significantly improving the predictive capability of the heat flux partitioning model concerning the CHF. The extended wall boiling model can be mathematically expressed as follows:

$$q_w = (q_{fc1} + q_{fc2} + q_q + q_{sl} + q_e)(1 - K_{dry}) + K_{dry} q_g \quad (19)$$

where K_{dry} denotes the wall dry-out area fraction. As indicated in Table 1, the definition of this parameter remains a subject of

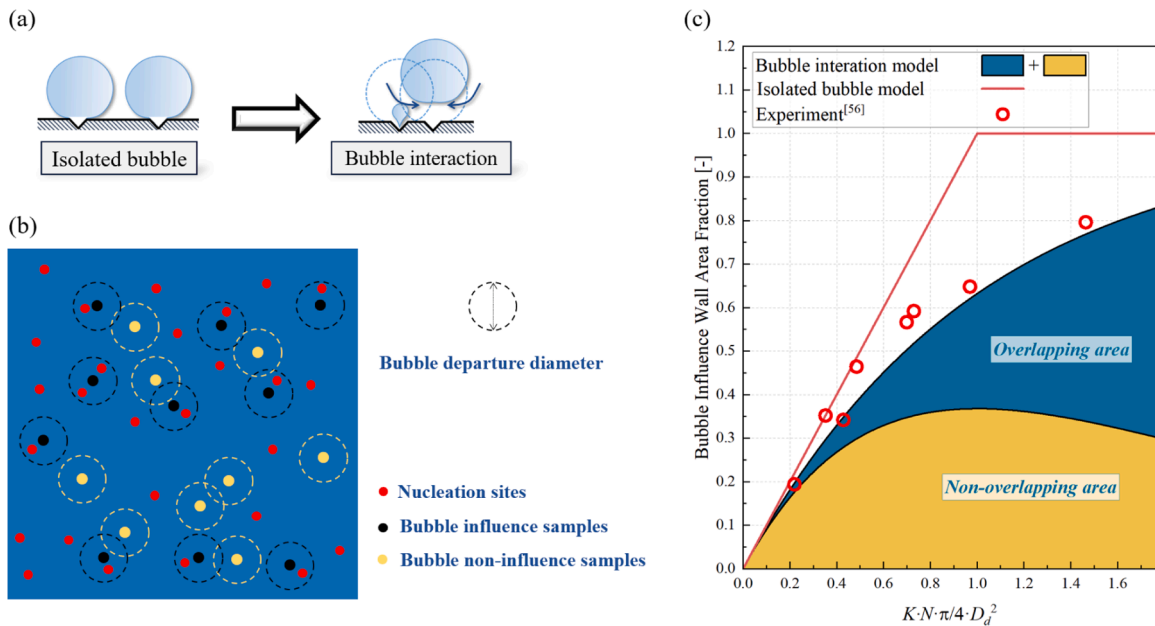


Fig. 3. Conceptual diagram of the bubble influence area: (a) Schematic representation of bubble interaction, (b) Quantitative analysis of bubble influence area overlap, (c) Validation of the bubble interaction model.

considerable debate within the existing literature. It is typically defined as a smooth phenomenological function. For example, Taherian [17] used a linear function related to the local bubble volume fraction.

$$K_{dry1} = \max\left(0, \min\left(\frac{\alpha_\delta - \alpha_{crit}}{0.95 - \alpha_{crit}}, 1\right)\right) \quad (20)$$

while Liang [34] employed a quadratic function, which can be expressed mathematically as:

$$K_{dry2} = \begin{cases} \frac{1}{2}e^{[-20(\alpha_{crit} - \alpha_\delta)]} & \alpha_\delta \leq \alpha_{crit} \\ 1 - \frac{1}{2}\left(\frac{1 - \alpha_\delta}{1 - \alpha_{crit}}\right)^2 & \alpha_\delta > \alpha_{crit} \end{cases} \quad (21)$$

In contrast, Liu [20,21] and Du [35] applied a cubic function as follows:

$$K_{dry3} = \begin{cases} 0 & \alpha_\delta \leq \alpha_{crit} \\ \left(\frac{\alpha_\delta - \alpha_{crit}}{1 - \alpha_{crit}}\right)^2 \left(3 - 2\frac{\alpha_\delta - \alpha_{crit}}{1 - \alpha_{crit}}\right) & \alpha_\delta > \alpha_{crit} \end{cases} \quad (22)$$

To facilitate subsequent comparative analysis, this study proposes the step function as follows:

$$K_{dry4} = \begin{cases} 0 & \alpha_\delta \leq \alpha_{crit} \\ 1 & \alpha_\delta > \alpha_{crit} \end{cases} \quad (23)$$

where α_δ represents the volume fraction of the bubble layer within a defined thickness (δ). α_{crit} denotes the critical void fraction, typically set at a value of 0.9.

Fig. 4 presents a comparative analysis of the definitions of wall dry-out area fraction as outlined in various studies. The different methodological approaches highlight the complexity and ongoing debate regarding the definition and representation of the wall dry-out area fraction in various theoretical frameworks.

Besides the definitions from phenomenological functions, some researchers have defined the wall dry-out area fraction using recent models related to the CHF mechanism. For example, Zhang [24,25] used the dry-patch model designed by Demarly [14] and Baglietto [57] to calculate the wall dry-out area fraction, which can be represented as follows:

$$K_{dry,Dry-patch} = \frac{A_{dry}}{A_{Heat}} = t_g f n_b \left(\frac{D_{dry}}{2}\right)^2 \quad (24)$$

where A_{dry} , A_{Heat} , and D_{dry} represent the dry-patch area, heated area, and dry-patch diameter, respectively.

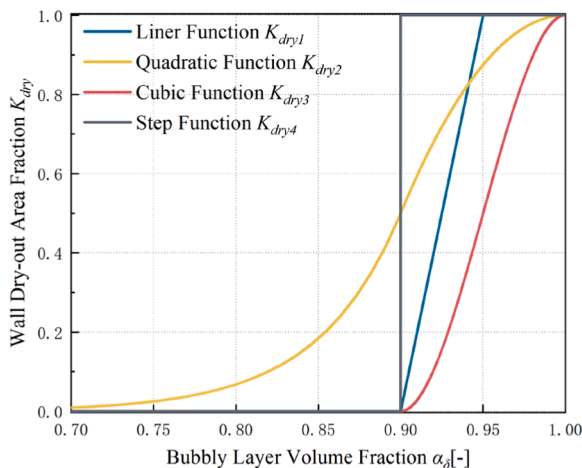


Fig. 4. Comparison of different wall dry-out area fraction definitions.

2.4.2. Quantitative comparison of various CHF criteria

The MIT series experiments [54] provided significant datasets regarding bubble dynamic parameters and CHF in a rectangular channel under various flow conditions. These datasets enabled refinement of the CHF criteria in the multiphase CFD approach. Table 2 presents the levels of each parameter considered in the experimental design. A full-factorial approach was employed, resulting in 27 experimental conditions.

The experiment loop features a rectangular test section (30 mm by 10 mm) with quartz windows for high-speed imaging on three sides and an infrared-opaque ITO heating module on the fourth side. It connects to a 965.2 mm entrance region. The flow field is initially solved by employing a single-phase solver, with the inlet velocity applied as a boundary condition for two-phase flow solvers. The experimental configuration, boundary conditions, and mesh sensitivity analysis are illustrated in Fig. 5. Detailed model validation is available in our prior publication [15]. Building upon that groundwork, which successfully predicted evaporation heat flux, this study aims to quantify the uncertainties associated with various CHF criteria.

Under the operational conditions of $P = 1.0$ bar, $G = 1000$ kg/(m²·s), and $\Delta T_{sub} = 10$ K, the results for wall superheat, wall dry-out area fraction, and bubbly layer volume fraction, utilizing various definitions of K_{dry} , are presented in Fig. 6. While the wall dry-out area fraction, defined through various phenomenological functions, effectively represents the physical phenomenon of boiling crisis, notable differences were observed in the patterns of wall temperature rise.

The wall dry-out area fraction increased more significantly with step and linear functions compared to quadratic and cubic functions. Notably, the quadratic function, K_{dry2} , showed the earliest increase in temperature. Further analysis of Figs. 4 and Fig. 6(b) indicates that K_{dry2} initiated heat transfer to the vapor even prior to the bubbly layer volume fraction reaching its critical value. This premature heat transfer interrupted the near-wall energy balance, resulting in an earlier rise in wall temperature. This observation implies that the initiation of wall temperature increases is closely associated with the specific value of the wall dry-out area fraction.

In addition, as illustrated in Fig. 6(b), the dry-spot-based wall dry-out area fraction remains at just 0.1, even when the wall heat flux reaches twice the experimentally determined CHF. At this moment, the boiling wall has not yet reached the threshold of energy imbalance. The primary contributor is the nucleate boiling heat transfer fraction ($1 - K_{dry}$), while a smaller portion of the heat flux is attributed to vapor convection. Consequently, the wall superheat is similar to the case with non-extending vapor convection ($K_{dry} = 0$).

As illustrated in Fig. 6(c), the comparison of non-extending vapor convection shows that different wall dry-out area fraction models do not influence the void fraction until a critical threshold of the bubbly layer volume fraction is reached. After surpassing this threshold, the wall dry-out area fraction leads to wall heat flux being allocated to vapor convection, necessitating a significant temperature difference for effective removal of the heat flux and resulting in a marked increase in wall temperature.

The wall dry-out area fraction significantly influences the boiling crisis phenomenon in multiphase CFD. A phenomenological function-based approach to defining this parameter is more effective than using the dry-spot CHF mechanism model. This phenomenological approach, which is determined by critical void fraction and bubble layer thickness,

Table 2

Experimental conditions for mass flux, pressure, and inlet subcooling in the MIT series.

Parameter	Level
Mass flux (G)	1000, 1500, 2000 kg/(m ² ·s)
Pressure (P)	1, 2, 5 bar
Inlet subcooling (ΔT_{sub})	5, 10, 20 K

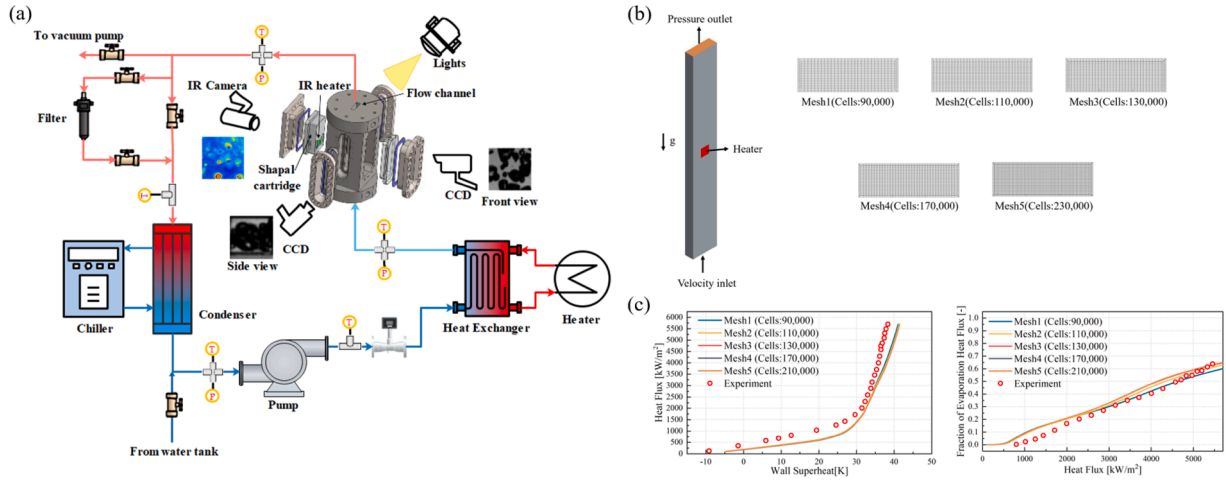


Fig. 5. The details of (a) Experiment setup and (b) Boundary conditions and (c) Mesh sensitivity analysis.

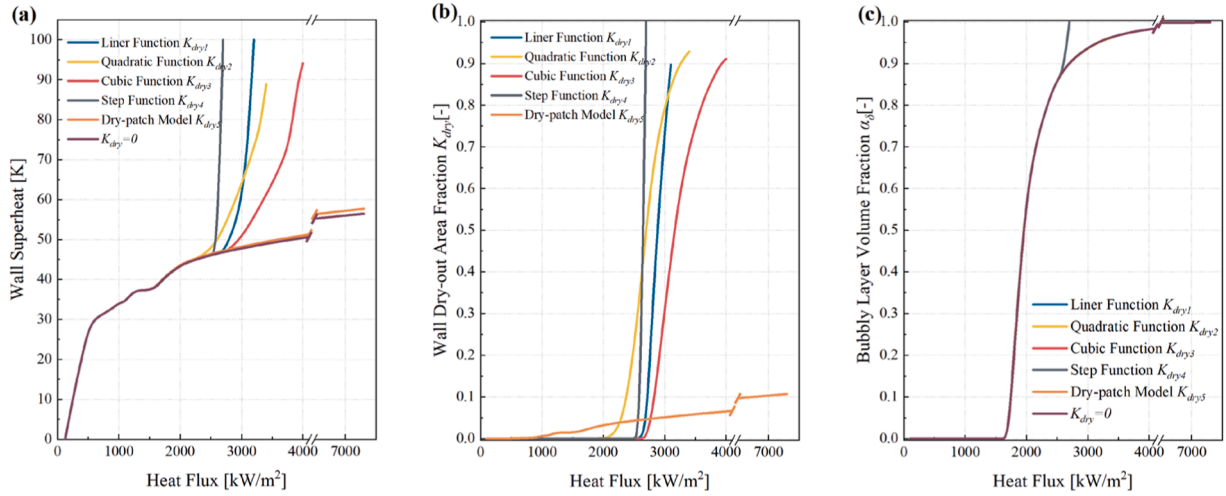


Fig. 6. Impact of wall dry-out area fraction definitions on numerical simulation of CHF phenomenon: (a) Wall superheat, (b) Wall dry-out area fraction, (c) Bubbly layer volume fraction.

aligns criteria based on wall temperature with those based on local volume fraction. Despite the importance of these critical thresholds, the development of a mechanistic model through physical experiments to address the associated bubble dynamics parameters remains a challenge.

This situation introduces considerable uncertainty into CFD-based predictions of CHF.

The comparison of predicted CHF with experimental results from MIT is illustrated in Fig. 7(a) and Fig. 7(b) across 27 different conditions.

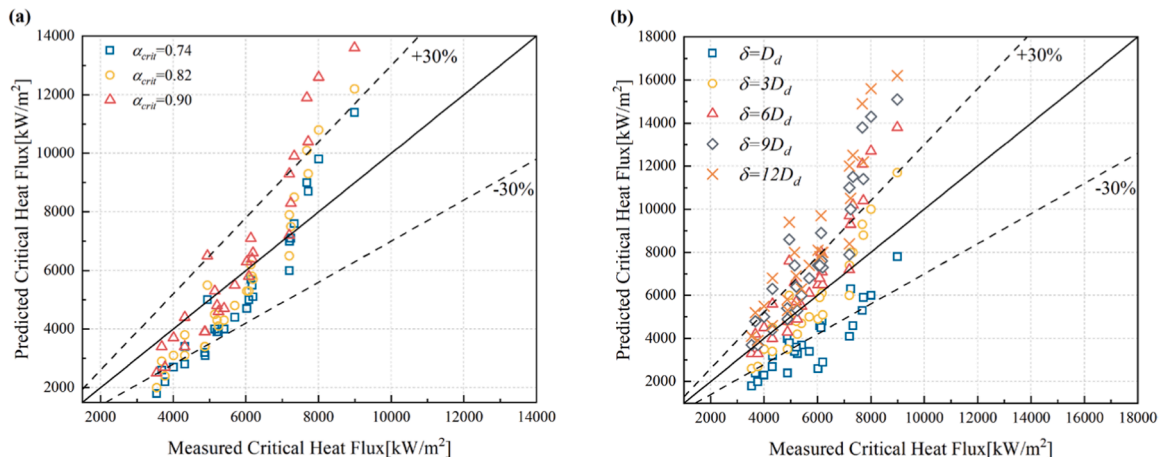


Fig. 7. Factors influencing CHF predictions: (a) Critical void fraction, (b) Bubbly layer thickness.

This analysis utilizes varying definitions of the critical void fraction and bubble layer thickness. In Fig. 7(a), the bubble layer thickness is defined as $y^+ = 250$. Additionally, in Fig. 7(b), the critical threshold for the void fraction is set at 0.82, while the bubble layer thickness is represented as an integer multiple of the bubble departure diameter (D_b). It is evident that these defined critical thresholds introduce substantial uncertainty into the CHF predictions within the multiphase CFD framework. Specifically, an increase in the critical void fraction and bubble layer thickness corresponds to higher predicted CHF values.

2.4.3. Deviation trends in CHF predictions based on Multiphase CFD

The multiphase CFD approach demonstrates significant sensitivity to critical thresholds, which necessitates the repeated calibration of associated parameters to ensure accurate predictions of the CHF, as demonstrated by Brewster [23]. However, such a redundant calibration process introduces substantial uncertainty. Does this imply a failure of the multiphase CFD method in predicting CHF?

Fig. 8 illustrates the deviations derived from the multiphase CFD analysis and MIT experiment across 27 operational conditions. The CHF criteria involved a critical void fraction of 0.82 and a bubble layer thickness set at $y^+ = 250$. Predictive deviations reveal a distinct trend: at low pressure and low mass flux, predictions generally underestimate the CHF, whereas at high pressure and high mass flux conditions, they tend to overestimate CHF. It is a straightforward task to determine whether the forthcoming predicted CHF value is above or below the corresponding experimental value.

In summary, the deviations between CFD predictions and experimental CHF values exhibit systematic patterns that are traceable to empirical assumptions in bubble dynamics models, such as fixed critical void fraction thresholds and uniform bubble layer thickness. These deviations inherently encode the limitations of current CFD approaches in capturing complex bubble interactions, thereby offering a learnable signal for neural networks. By training on the residuals derived from CFD, the proposed framework adjusts the relationship between input parameters and CHF. This approach effectively reduces the reliance on manual calibration of α_{crit} and δ . Furthermore, integrating the physical principles from CFD, such as the Eulerian-Eulerian two-fluid model and wall boiling sub-models, not only improves predictive accuracy but also allows for extrapolation to untested conditions, as validated in Fig. 16.

3. Data acquisition, preprocessing, and multiphase CFD-informed NN

The multiphase CFD framework offers valuable insights into the mechanisms underlying CHF; however, its reliance on empirical

thresholds introduces systematic biases, as illustrated in Fig. 8. These deviations exhibit discernible patterns associated with uncalibrated parameters governing bubble dynamics. To address this limitation, we propose the development of a Multiphase CFD-informed Neural Network (CFD-informed NN for short) that effectively learns the deviations between CFD predictions and experimental CHF data. By incorporating the CFD model as foundational domain knowledge, the proposed network compensates for modeling inadequacies while maintaining adherence to physical consistency. The following section will detail the architecture of this CFD-informed NN and outline its training strategy.

3.1. CHF dataset and standardization

To effectively validate the performance of the CFD-informed NN, it is crucial to construct a CHF dataset that encompasses training, validation, and test sets. Given the scarcity of operational data from the MIT boiling flow experiments, which consist of only 27 data points, this study leverages the 2006 CHF lookup table [58] to define the target CHF values. The dataset focuses on a vertical upward tube with a diameter of 8 mm and a length of 1 m. The selected operating conditions cover a pressure range of 7 MPa to 18 MPa, mass velocities ranging from 750 to 5000 kg/(m²·s), and exit qualities from -0.2 to 0.25, resulting in a total of 600 data points. It is important to recognize that the 2006 CHF lookup table provides only the local quality of the outlet, lacking any accompanying data concerning the inlet conditions. Based on the principle of energy conservation, it is possible to determine the inlet enthalpy and quality as follows:

$$h_{in} = h_{out} - \frac{4q_{CHF}L}{GD} \quad (25)$$

$$x_{in} = x_{out} - \frac{4q_{CHF}L}{GDh_{fg}} \quad (26)$$

where h_{in} and h_{out} represent the specific enthalpies at the inlet and outlet. x_{in} and x_{out} denote the qualities at the inlet and outlet, respectively. The tube length L is specified as 1 m, while the diameter D is set at 8 mm. The value of q_{CHF} is derived from the CHF lookup table. Following the exclusion of data points characterized by an inlet enthalpy of less than zero and a quality greater than zero, the refined database comprises a total of 433 data points. Further detailed information is available in the *Supplementary Material*. Notably, for high-pressure applications (7–18 MPa), we adopted industry-validated submodels supported by prior literature [24,36] to address pressure-dependent boiling phenomena. Specifically, the nucleation site density was modeled using the Li

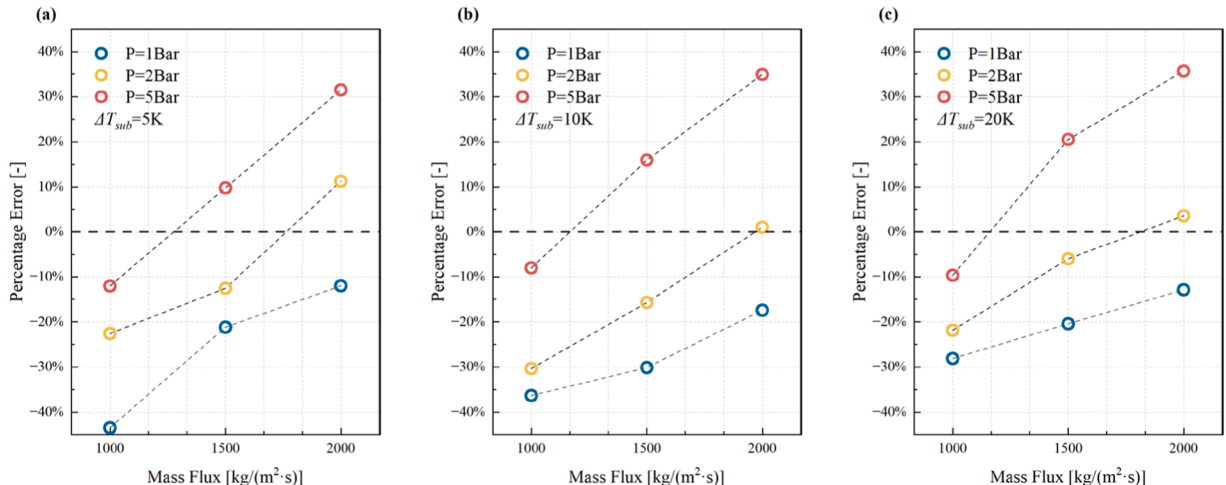


Fig. 8. Regularity in CFD prediction deviations: (a) $\Delta T_{sub} = 5$ K, (b) $\Delta T_{sub} = 10$ K, (c) $\Delta T_{sub} = 20$ K.

correlation [59], the bubble departure diameter via the Kocamustafaoğullari correlation [60], and the bubble departure frequency through the Cole correlation [61].

The CFD-predicted CHF values were determined based on the bubble crowding criterion, i.e., the boiling crisis is triggered when the local void fraction within the bubbly layer adjacent to the heated wall reaches the critical threshold of 0.82, consistent with the theoretical framework proposed by Weisman and Pei [1].

In this study, the input features consist of pressure, mass flux, and inlet subcooling, which exhibit substantial dimensional disparities. To standardize these features and ensure equitable treatment in the machine learning pipeline, we utilized the Z-Score normalization method. This method transforms each feature to have a mean of zero and a standard deviation of one, enhancing the compatibility of the input data with various machine-learning algorithms. The standardization of each input feature can be mathematically represented as follows:

$$Z = \frac{(x - \mu)}{\sigma} \quad (27)$$

where x represents an individual data point from the raw dataset of each input feature. μ denotes the arithmetic mean of each input feature, while σ represents the standard deviation of each input feature.

The dataset was divided before applying the standardization transform to prevent unintentional information leakage from the test set. Specifically, the shuffled dataset was split into training, validation, and test sets in the proportions of 80 %, 10 %, and 10 %, respectively. After standardizing the training set, the calculated mean and standard deviation were applied to transform both the validation and test sets.

3.2. Multiphase CFD-informed neural network building

Both the CFD-informed NN and the traditional standalone neural network (or multilayer perceptron, Standalone NN for short) were trained and validated using the PyTorch framework and the scikit-learn library in Python 3.12. The construction of the neural network involved designing the network structure, implementing forward propagation, and performing backpropagation to update the weights and biases of each neuron in the network.

3.2.1. Model structure

The CFD-informed NN is structured similarly to a Physics-informed machine learning-aided framework, with a primary distinction being its incorporation of a prior model derived from CFD. Fig. 9 delineates the workflow of the CFD-informed NN during the stages of training and validation/testing. The input features utilized in this framework encompass pressure (P), mass flux (G), and inlet subcooling (ΔT_{sub}), all of which are also necessary parameters for performing CFD analyses.

During the training stage, residuals (ϵ) are determined by subtracting the values predicted by a multiphase CFD model (y_p) from the corresponding target values derived from the 2006 CHF lookup table (y). Subsequently, a deep feedforward neural network (DFNN) is trained using these residuals (ϵ). The training procedure incorporates backpropagation, a process in which the network's weights and biases are iteratively refined to minimize the deviation between the residuals produced by the neural network (ϵ_m) and the target residuals (ϵ). This deviation is quantitatively evaluated through a designated loss function. The well-trained CFD-informed NN produces outputs that consist of the sum of the predicted values from the CFD model (y_p) and the deviations predicted by the DFNN (ϵ_m).

3.2.2. Training and validation

The training process of a neural network involves several fundamental steps, namely forward propagation, loss computation, backpropagation, and subsequent parameter updates. Each hidden layer employs the ReLU activation function to introduce non-linearity,

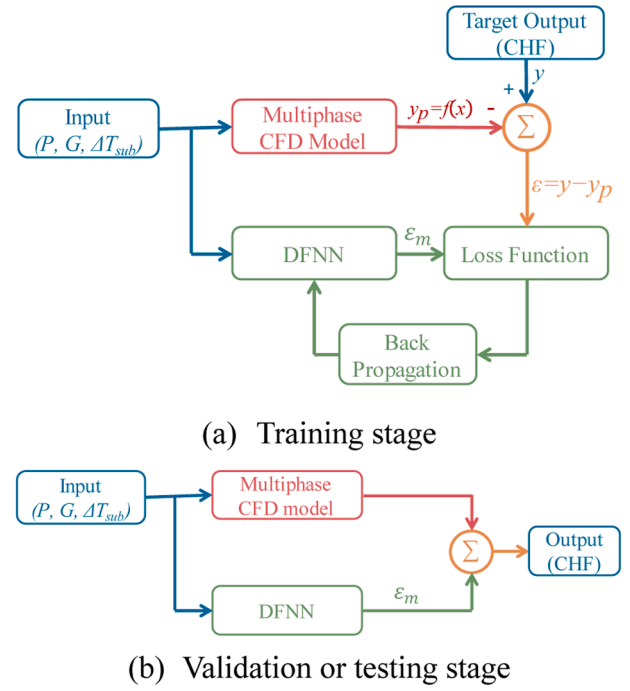


Fig. 9. Structure for Multiphase CFD-informed NN.

enabling the model to capture complex data patterns. To address vanishing and exploding gradient issues, the Kaiming initialization method is employed for initializing neuron weights, which is expected to enhance training efficiency and overall performance.

The loss function employed in this study is the Mean Squared Error (MSE), which quantifies the discrepancy between the predicted and target values. In the pursuit of minimizing the loss function, the Adam optimizer, known for its adaptive moment estimation capabilities, is employed. To enhance the model's generalization to unseen data and mitigate the risk of overfitting, strategies such as early stopping and learning rate decay are implemented during the training process. The evaluation of predictive performance for the model is ultimately conducted through the application of the Root Mean Squared Error (RMSE) and the determination coefficient (R^2). The mathematical formulations for these metrics are as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (\varphi_{i,cal} - \varphi_{i,exp})^2 \quad (28)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\varphi_{i,cal} - \varphi_{i,exp})^2} \quad (29)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (\varphi_{i,exp} - \varphi_{i,cal})^2}{\sum_{i=1}^n (\varphi_{i,cal} - \bar{\varphi}_{i,cal})^2} \quad (30)$$

3.2.3. Bayesian-based hyperparameter optimization

Bayesian optimization is an advanced global optimization methodology that leverages probabilistic models to enhance the efficiency of the optimization process. It constructs a posterior probability model of the objective function, which is subsequently refined through the incorporation of actual observations. This dynamic updating enables the identification of regions within the search space that are most likely to contain the global optimum. Consequently, this framework facilitates the determination of optimal hyperparameter configurations while significantly reducing the number of sampling iterations required. The inherent efficiency of Bayesian optimization makes it particularly

valuable in scenarios where evaluations of the objective function are expensive or time-consuming.

The flowchart depicting the Bayesian optimization process is presented in Fig. 10. This research employs the average loss function derived from 10-fold cross-validation as the objective function corresponding to each combination of hyperparameters. Following the acquisition of the objective function, a Gaussian process is utilized as a surrogate model to approximate its behavior. Furthermore, an acquisition function is employed with the Tree-structured Parzen Estimator (TPE) global optimization algorithm to identify the next most promising combination of hyperparameters, aiming to minimize the objective function.

The maximum allowable iterations for the Bayesian optimization process are limited to 200. Fig. 11 presents the evolution of the optimal objective function value, defined as the loss function obtained through 10-fold cross-validation, across the entire optimization process. Notably, there is a marked decrease in the optimal objective function value during the initial 25 iterations. Subsequently, the rate of change diminishes, indicating a gradual convergence toward a stable solution. Additionally, Table 3 provides a detailed summary of the search range for the hyperparameters alongside the optimal values performed through Bayesian optimization. It is important to emphasize that the execution of Bayesian optimization should be confined exclusively to the training and validation sets in order to preserve the independence and integrity of the test set.

4. Results and discussion

4.1. Model generalization performance: CFD-informed NN vs. standalone NN

During the process of model training, the weights and biases of each neuron are updated iteratively, employing hyperparameter configurations identified through Bayesian optimization, with the overarching objective of minimizing the loss function. This research evaluates the model's generalization capabilities from two distinct perspectives: interpolation and extrapolation. For the assessment of interpolation, the dataset is shuffled, and a random subset comprising 10 % of the data is selected as the test set. In contrast, the extrapolation assessment involves establishing a cut-off pressure value. Data points exhibiting pressures

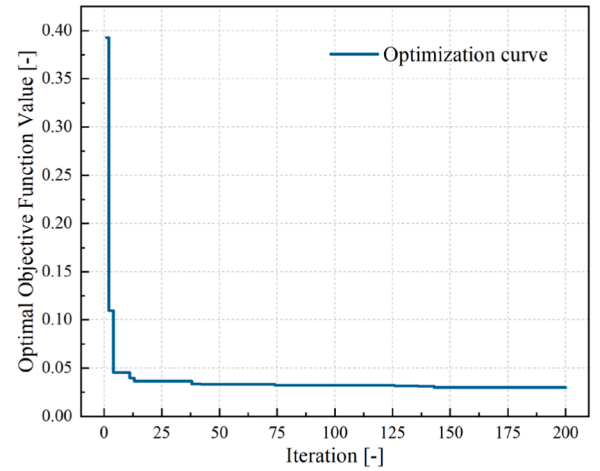


Fig. 11. Performance of Bayesian optimization over iterations.

Table 3

Bayesian-based hyperparameter optimization: Search space and search results.

Hyperparameter	Search Range	Sampling Method	Best Value
The depth of hidden layers	3–6	Discrete uniform distribution	3
The number of hidden layer nodes	1–128	Discrete uniform distribution	94, 69, 78
Initial learning rate	$1e^{-6}$ – $1e^{-2}$	Continuous Uniform on Log Scale	$7.8e^{-5}$
Batch size	16, 32, 64, 128	Discrete choice	32
Epochs	100–1000	Discrete uniform distribution	798

below this threshold are allocated to the training and validation sets, whereas those exceeding the threshold are classified as the test set. Throughout the training process, it is imperative to ensure that the test set remains unobserved by the model, thereby preserving the integrity of the evaluation process.

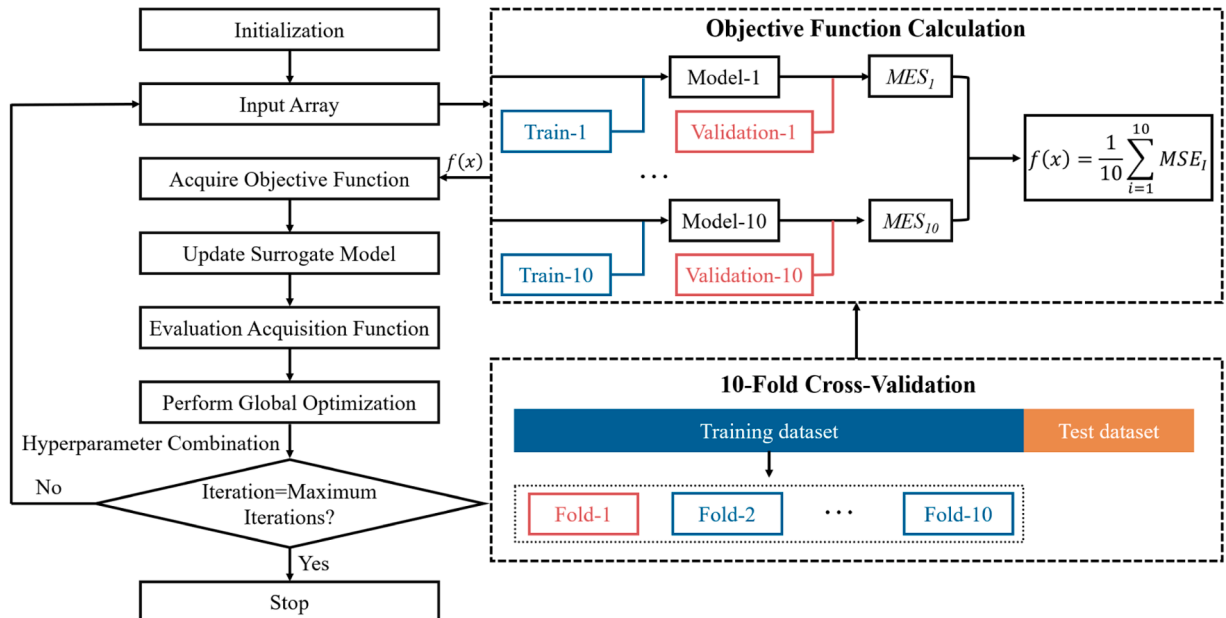


Fig. 10. Bayesian-based hyperparameter optimization flowchart.

Interpolation performance

Fig. 12 presents a parity plot comparing the model's predicted values to the target values from the test set during the interpolation evaluation. The CFD-informed NN showed excellent predictive performance, achieving an RMSE of 0.119 and an R^2 of 0.993. In comparison, the data-driven standalone NN exhibited strong predictive capabilities as well, with an RMSE of 0.269 and an R^2 of 0.962. However, it did show significant prediction deviations at certain operating conditions.

In order to conduct a comprehensive analysis and comparison of the performance metrics of the two models, Fig. 13 presents an Error Cumulative Distribution Function (ECDF) plot, and Fig. 14 delineates the proportion of data with absolute relative errors of 5 %, 10 %, and 20 %. A rigorous statistical analysis was performed to quantify the precision of the model's predictions. The CFD-informed NN achieved a relative prediction error of 10 % for 95 % of the test set, indicating high predictive accuracy. In contrast, the Standalone NN exhibited comparatively lower predictive accuracy, with only 73 % of the test set falling within the same relative prediction error threshold of 10 %. It is important to note that although the data-driven Standalone NN demonstrated lower accuracy compared to the CFD-informed NN, it still exhibited significantly enhanced interpolation capabilities than conventional CFD approaches.

Extrapolation performance

In the context of practical deployment in engineering applications, the capability of a model to extrapolate beyond the training data is frequently of greater importance than its ability to interpolate within the specified data range. Conducting CHF experiments at elevated pressures presents substantial resource demands and technical challenges. Conversely, CHF data obtained at lower pressures are more readily accessible and extensively documented. Consequently, this study aims to assess the model's performance regarding its ability to extrapolate data beyond the pressure range on which it was originally trained.

The dataset was partitioned into two distinct groups based on a predetermined pressure cutoff threshold (P_{cutoff}). Data points exhibiting pressures below this threshold were utilized for the training and validation of the predictive model. Conversely, data points with pressures equal to or exceeding the threshold were allocated as the test set, which served to assess the model's extrapolation capabilities. This study systematically examined all potential pressure cutoff thresholds within the range encompassed by the dataset. Table 4 delineates the number of instances within the training and test sets across various pressure cutoff thresholds, as well as the predictive performance of two models evaluated under these conditions. This performance is quantitatively represented through metrics such as RMSE and R^2 .

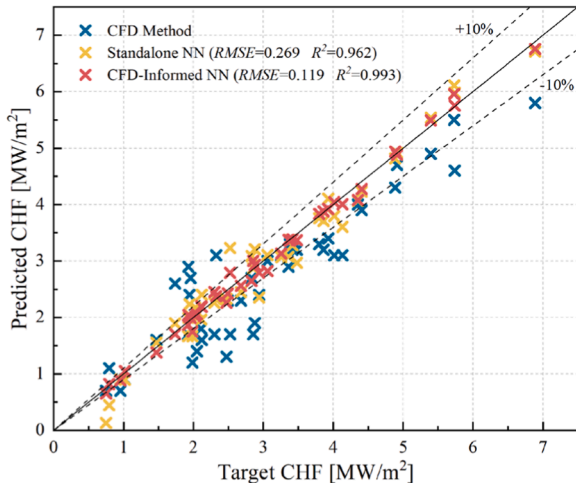


Fig. 12. Measured vs. predicted parity plot.

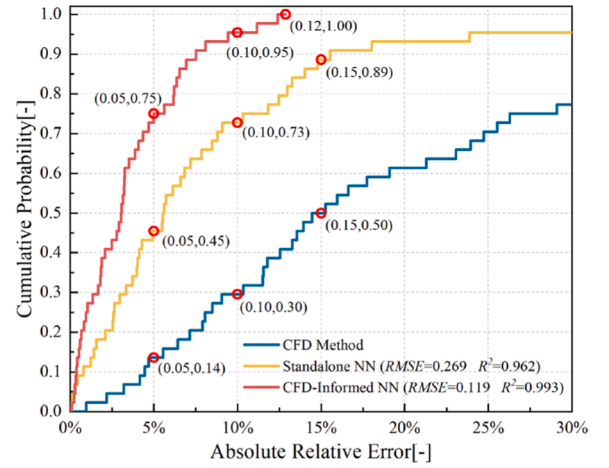


Fig. 13. Error cumulative distribution function plot.

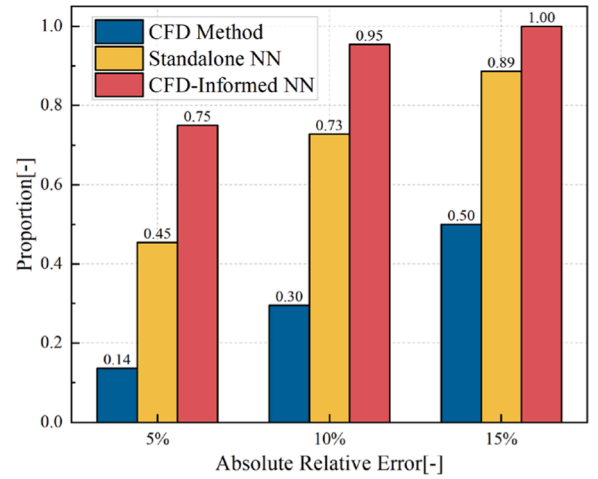


Fig. 14. Proportion of data with absolute relative error of 5 %, 10 %, and 20 %.

Table 4

Model performance on extrapolation to higher pressures.

Cutoff Pressure P_{cutoff} [MPa]		10	12	14	16	18
No. of training data		50	108	177	251	337
No. of test data		383	325	256	182	96
Test data	CFD-informed	0.392	0.339	0.272	0.228	0.181
RMSE	NN					
Test data R^2	Standalone NN	5.141	2.819	0.860	0.492	0.347
	CFD-informed	0.937	0.941	0.948	0.955	0.955
	NN					
	Standalone NN	-9.673	3.094	0.488	0.791	0.834

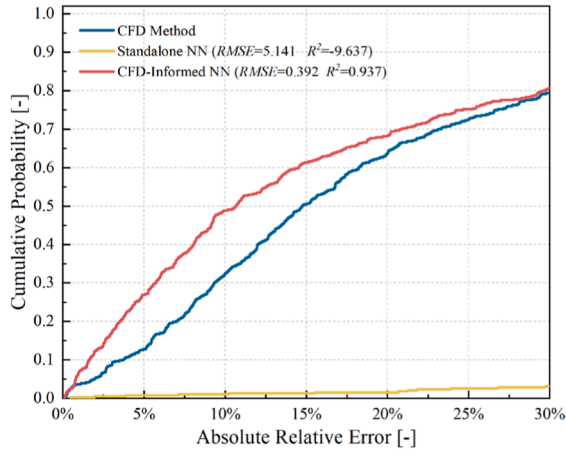
The assessment of the Standalone NN revealed a significant limitation in its extrapolation capabilities. Specifically, when the pressure cutoff threshold was established at 18 MPa, the model was trained and validated using 337 data points within the 7–16 MPa range and subsequently tested with an additional 96 data points at 18 MPa. The results yielded an RMSE of 0.347 and R^2 of 0.834. These findings underscore the necessity for caution in the application of the Standalone NN beyond the training data range in practical scenarios.

In contrast, the CFD-informed NN demonstrated markedly enhanced extrapolation capabilities. With a pressure cutoff threshold set at 10 MPa, this model was trained and validated with 50 data points at 7 MPa, followed by testing with 383 data points spanning the 10–18 MPa range. The RMSE recorded was 0.392, while the R^2 improved significantly to

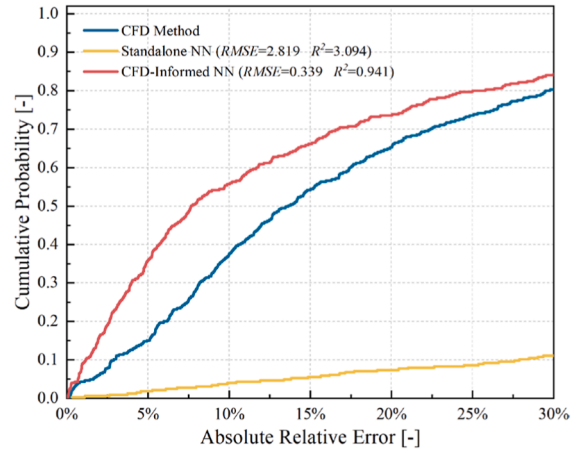
0.937. This substantial improvement in predictive accuracy indicates that the CFD-informed NN is more adept in functioning across a wider pressure range compared to the Standalone NN.

Fig. 15 presents the cumulative distribution functions of errors

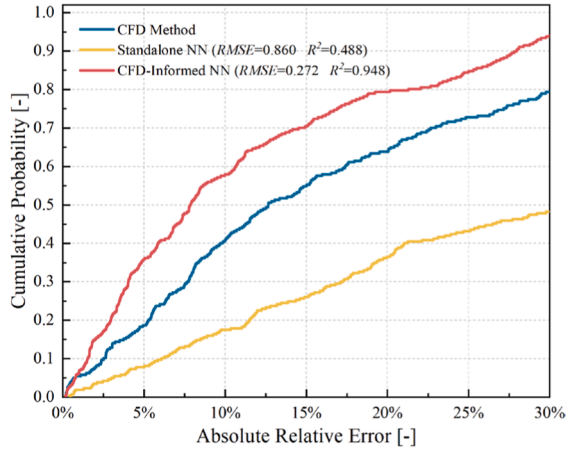
associated with both models across various pressure cutoff thresholds (P_{cutoff}). At a pressure cutoff of 16 MPa, the data-driven-based Standalone NN demonstrated extrapolative generalization accuracy that was comparable to the traditional CFD method when evaluated against



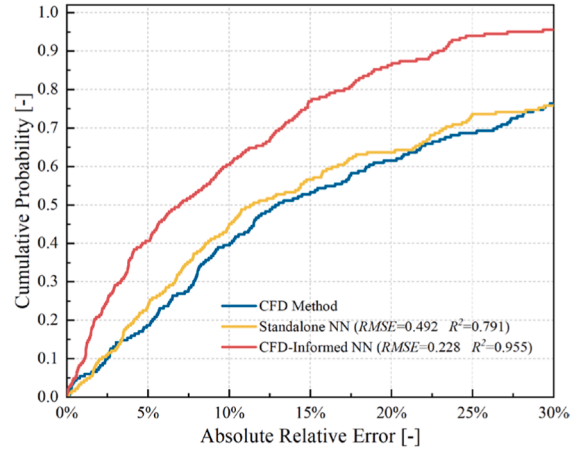
(a) $P_{cutoff}=10$ MPa



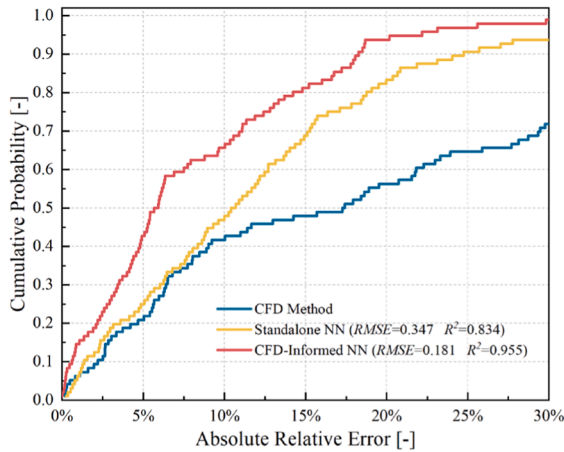
(b) $P_{cutoff}=12$ MPa



(c) $P_{cutoff}=14$ MPa



(d) $P_{cutoff}=16$ MPa



(e) $P_{cutoff}=18$ MPa

Fig. 15. Accumulated error function plots under various pressure cutoff thresholds.

unseen data. In contrast, when the pressure cutoff was decreased to 10 MPa, the CFD-informed NN also exhibited significant predictive capabilities, with 80 % of the relative errors remaining within 30 %.

An analysis was conducted using a pressure cutoff threshold of 14 MPa, with a dataset of 256 data points representing pressures at 14 MPa, 16 MPa, and 18 MPa as the test dataset. The relative errors for this test set are illustrated in Fig. 16 using violin plots. The results reveal that the relative error for the Standalone NN experiences a rapid increase as the pressure is extrapolated beyond the training range. Conversely, the CFD-informed NN demonstrates a prediction bias that remains within acceptable limits even when extrapolated to a pressure of 18 MPa.

The extrapolation analysis was systematically extended to evaluate mass flux and inlet subcooling parameters, as presented in Tables 5 and Table 6. This comprehensive evaluation demonstrated consistent performance advantages of the CFD-informed NN across all operational dimensions. According to Table 5, at a mass flux cutoff of 2500 kg/(m²·s) and utilizing only 143 training samples, the CFD-informed NN achieves superior generalization metrics ($RMSE = 0.357$, $R^2 = 0.956$) compared to the standalone NN ($RMSE = 1.187$, $R^2 = 0.522$). This significant reduction in error, even in data-limited scenarios, underscores the framework's ability to preserve the physical relationships that are encoded in the CFD model's domain knowledge. Similar patterns are evident in the inlet subcooling extrapolation (Table 6), particularly under challenging conditions at a 50 K cutoff. In this case, the model reveals robust performance ($R^2 = 0.955$) despite being trained on just 134 low-subcooling samples. This extremely training-to-test ratio confirms the framework's capacity to maintain physical consistency, even with considerable data imbalances.

These findings highlight that the extrapolation and generalization capabilities of the CFD-informed NN significantly surpass those of the Standalone NN. This enhanced performance suggests that the CFD-informed NN holds considerable potential for enhancing future CHF prediction tools designed for thermodynamic system applications.

4.2. Interpretability of the CFD-informed NN

The CFD-informed NN demonstrates enhanced generalization capabilities, particularly in extrapolation, when compared to the Standalone NN. This advantage derives from the CFD-informed NN's ability to learn the deviation between predictions generated by CFD simulations and the target values. The networks leverage a comprehensive multiphase CFD program that incorporates a variety of physical models, including the Eulerian-Eulerian two-fluid model, the wall boiling model, and the bubble dynamics model, to establish a nonlinear mapping between input

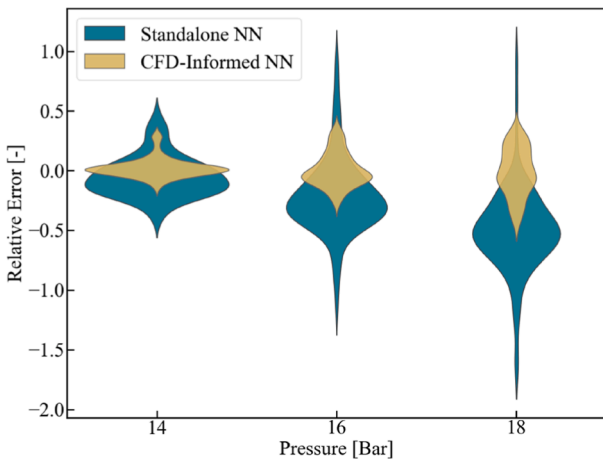


Fig. 16. Violin plots of relative error for standalone NN and CFD-informed NN at $P_{cutoff}=14$ MPa.

Table 5

Model performance on extrapolation to higher mass fluxes.

Cutoff Mass Flux [kg/(m ² ·s)]		2500	3000	3500	4000	4500
No. of training data		143	193	242	291	340
No. of test data		290	240	191	142	93
Test data	CFD-informed	0.357	0.311	0.255	0.248	0.201
RMSE	NN					
Test data R ²	Standalone NN	1.187	0.827	0.782	0.566	0.384
	CFD-informed	0.956	0.968	0.980	0.982	0.988
	NN					
	Standalone NN	0.522	0.780	0.816	0.907	0.958

Table 6

Model performance on extrapolation to higher inlet subcoolings.

Cutoff Inlet subcooling[K]		50	100	150	200	250
No. of training data		134	228	299	353	394
No. of test data		299	205	134	80	39
Test data	CFD-informed	0.346	0.244	0.211	0.207	0.207
RMSE	NN					
Test data R ²	Standalone NN	1.174	1.088	0.983	0.742	0.598
	CFD-informed	0.955	0.979	0.987	0.989	0.990
	NN					
	Standalone NN	0.493	0.603	0.720	0.864	0.922

features and CHF.

During this mapping process, the effects of local parameters relevant to CHF are highlighted. These include the local two-phase flow field and the local void fraction within the bubble layer. Acquiring an understanding of these parameters is essential for effectively addressing the boiling crisis; however, they present considerable challenges for direct measurement in empirical experiments. This limitation restricts the Standalone NN's ability to incorporate these essential variables into its predictive model.

Fig. 17 presents the probability density distributions of the original Look-up table values, which function as the reference targets for the Standalone NN, alongside the deviations from CFD predictions, serving as the targets for the CFD-informed NN within the interpolation test set. It is noteworthy that the skewness (S_{ke}) of the CFD prediction deviations approaches zero, which indicates a relatively symmetric distribution when compared to the original Look-up table values. Furthermore, the kurtosis (K_{ur}) of the CFD prediction deviations is also approximately zero, suggesting that the distribution exhibits a degree of sharpness or flatness reminiscent of a normal distribution.

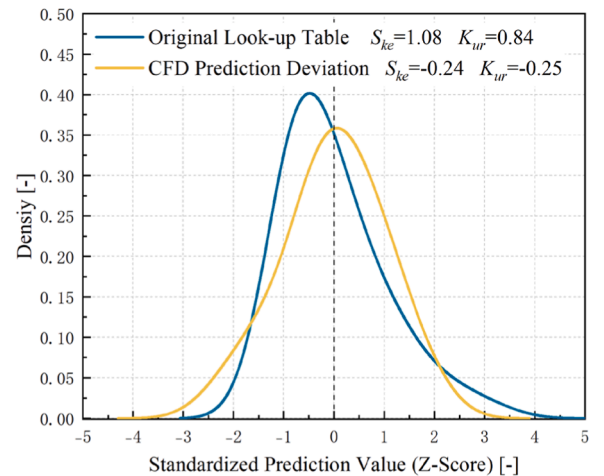


Fig. 17. Probability density distribution of standardized prediction values (Z-Scores).

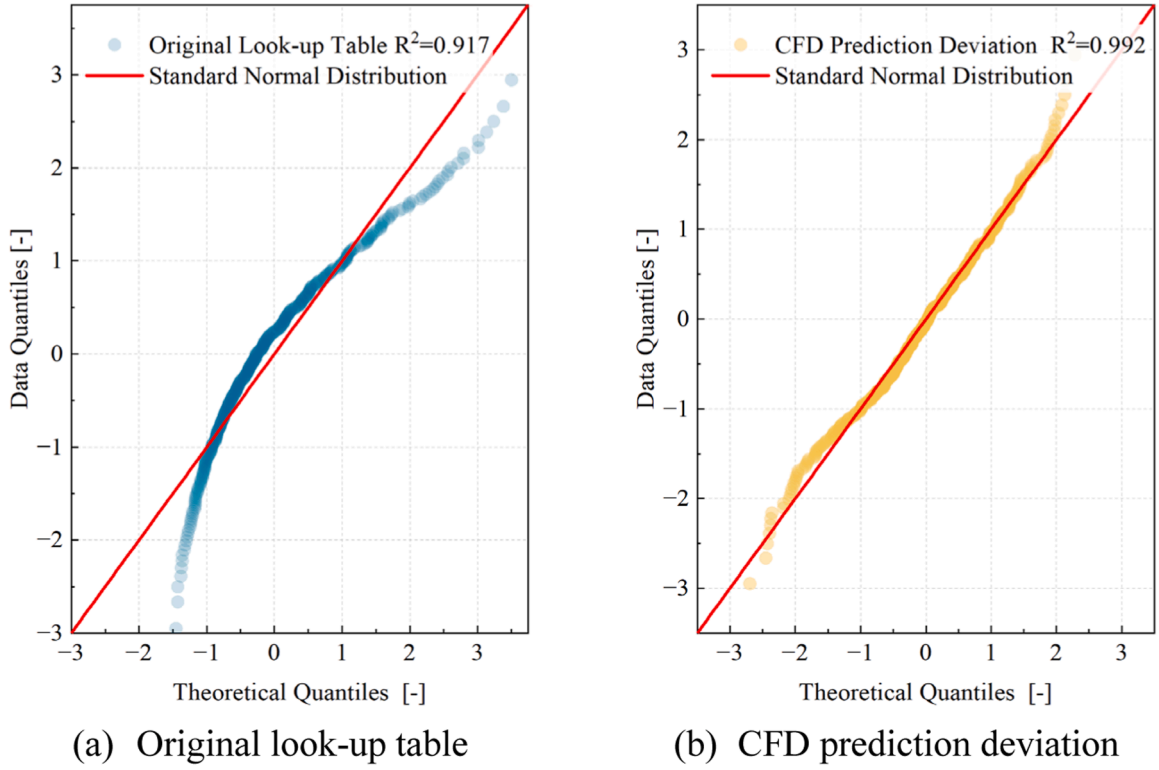


Fig. 18. Comparative quantile-quantile analysis: original look-up table vs. CFD prediction deviation.

Fig. 18 presents the quantile-quantile (Q-Q) plot comparing CFD prediction deviations with values derived from the original look-up table within the interpolation test set. An analysis of the data reveals that the CFD prediction deviations exhibit a more pronounced linear relationship and demonstrate closer alignment with a normal distribution compared to the original look-up table values. The feedforward neural network, which is part of the CFD-informed neural network framework, leverages the detailed physical models provided by multiphase CFD simulations. This integration enables the network to effectively identify target values that display consistent patterns, thereby enhancing its capability to learn from and make predictions that are grounded in the foundational principles of physics. Consequently, the CFD-informed NN exhibits enhanced generalization capabilities, particularly in extrapolation tasks, compared to the standalone neural network.

4.3. Comparative prior model performance: CFD vs. empirical correlation

Indeed, empirical correlations also serve as essential prior models, facilitating intricate, physics-based mapping relationships. This study conducts a thorough comparative analysis of the interpolation performance between a PIML approach that learns deviations based on the established Tong [9] and W-3 [10] correlations, and the CFD-informed NN.

Fig. 19 illustrates the probability density distributions of the prediction deviations associated with the Tong [9], W-3 [10] correlations, and CFD within the test set. All three distributions demonstrate characteristics consistent with a normal distribution. A quantitative analysis reveals that the prediction deviations of the W-3 correlation [10] exhibit the lowest skewness among the three, indicating a more symmetric distribution. Conversely, the CFD prediction deviations have a significantly lower kurtosis ($K_{ur} = -0.25$), suggesting a flatter distribution relative to the prediction deviations observed in both the Tong and W-3 correlations.

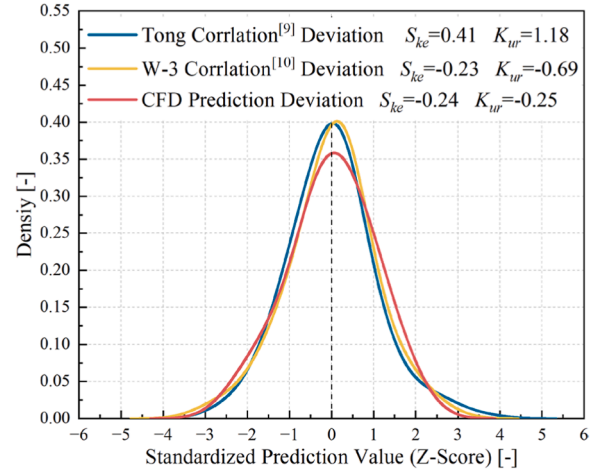


Fig. 19. Standardized prediction deviation probability densities across Tong, W-3 Correlation, and CFD.

Fig. 20 presents a comparative analysis of the quantile-quantile plots for the Tong and W-3 correlations. Following the application of nonlinear transformations to the input features based on empirical correlations, the resulting target values supplied to the DFNN demonstrate an enhanced degree of regularity compared to the original look-up table data. Specifically, the R^2 for the Tong and W-3 correlations is reported as 0.976 and 0.982, respectively, while the R^2 for the unprocessed original look-up table is recorded at 0.917. Remarkably, advancements in multiphase CFD models over recent years have culminated in prediction deviations that conform more closely to a normal distribution, achieving an R^2 of 0.992, which represents a notable improvement over the simple empirical correlations.

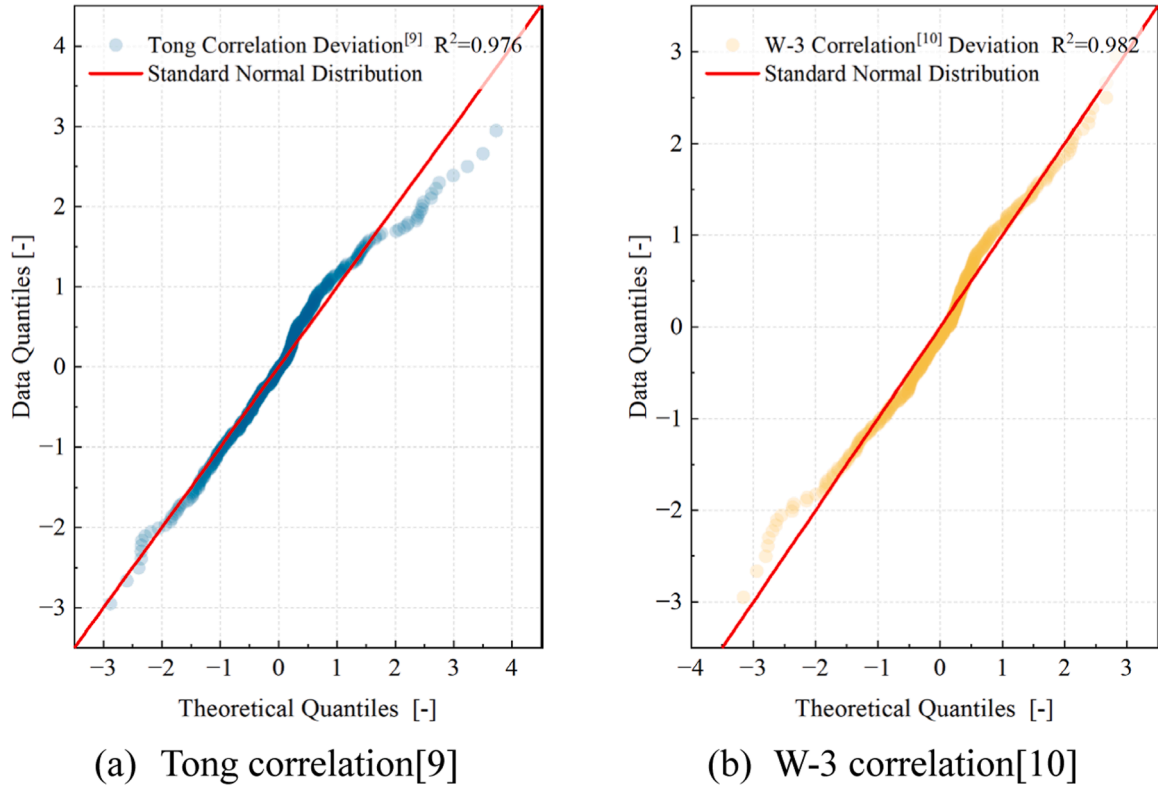


Fig. 20. Comparative quantile-quantile analysis: Tong correlation vs. W-3 correlation.

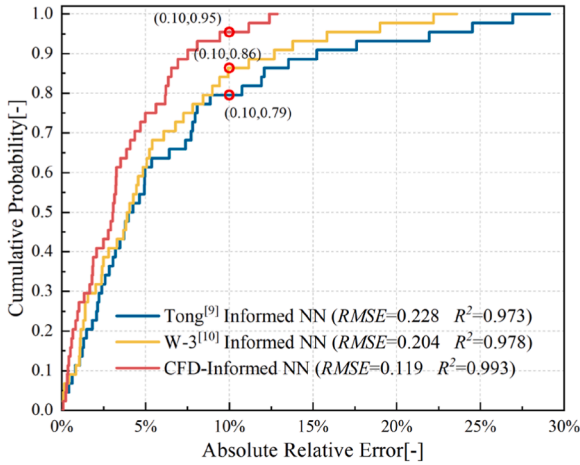


Fig. 21. ECDF comparison: CFD-informed NN, W-3 informed NN and Tong informed NN.

Fig. 21 highlights the crucial role of regular data distribution in enhancing the generalization capabilities of the model when evaluated against the test set. Notably, the physical insights derived from the CFD model significantly enhance the DFNN ability to capture data characteristics. Specifically, 95 % of the test data showed prediction deviations of within 10 %. In comparison, the W-3 informed Neural Network achieved 86 % of the test data with deviations within 10 %, while the Tong informed Neural Network exhibited a slightly lower accuracy, with 79 % of the test data showing deviations within the same error thread.

This study evaluates the extrapolation capabilities of three neural network architectures: the CFD-informed NN, the Tong informed NN, and the W-3 informed NN. To rigorously assess generalization beyond training domains, a pressure cutoff threshold of 14 MPa was

implemented. Data points below P_{cutoff} (177 samples, 7–12 MPa) comprised the training/validation set, while points above (256 samples, 14–18 MPa) formed the test set, ensuring no pressure overlap between sets.

Fig. 22 presents SHAP feature attribution maps for each framework. Normalized Shapley values (horizontal axis) quantify feature importance, with point colors indicating relative feature magnitudes. In the CFD-informed NN, pressure emerges as the dominant feature, showing a strong correlation between higher pressure (indicated by red points) and negative Shapley values. This observation is consistent with the CFD model's physical trend, which predicts reduced deviations in CHF predictions at higher pressures.

In contrast, the Tong-informed NN and W-3-informed NN reveal inconsistent relationships between pressure and Shapley values, lacking a coherent trend. Instead, inlet subcooling takes precedence in the corrections, highlighting a limitation in accurately extrapolating unseen pressure conditions. This underscores the challenges associated with empirical correlations in effectively capturing the complex mechanisms related to CHF.

Fig. 23 compares the Cumulative Distribution Functions of errors for three different frameworks using the 14–18 MPa test set. The CFD-informed NN achieves an RMSE of 0.272, demonstrating superior extrapolation capabilities compared to the Tong-informed NN, which has an RMSE of 0.603, and the W-3-informed NN, which has an RMSE of 0.724. Notably, 80 % of the CFD-informed NN's predictions have relative errors of <20 %. In contrast, over 53 % of the Tong-informed NN's test data exceeds a 20 % error, while the W-3-informed NN has >67 % of its test data surpassing this threshold. This significant difference highlights the CFD framework's ability to regularize residuals through physics-based feature transformations, thus enabling more robust extrapolation.

Overall, the comprehensive insights provided by the multiphase CFD framework, encompassing local flow field and void fraction, significantly enhance the understanding of underlying physical phenomena

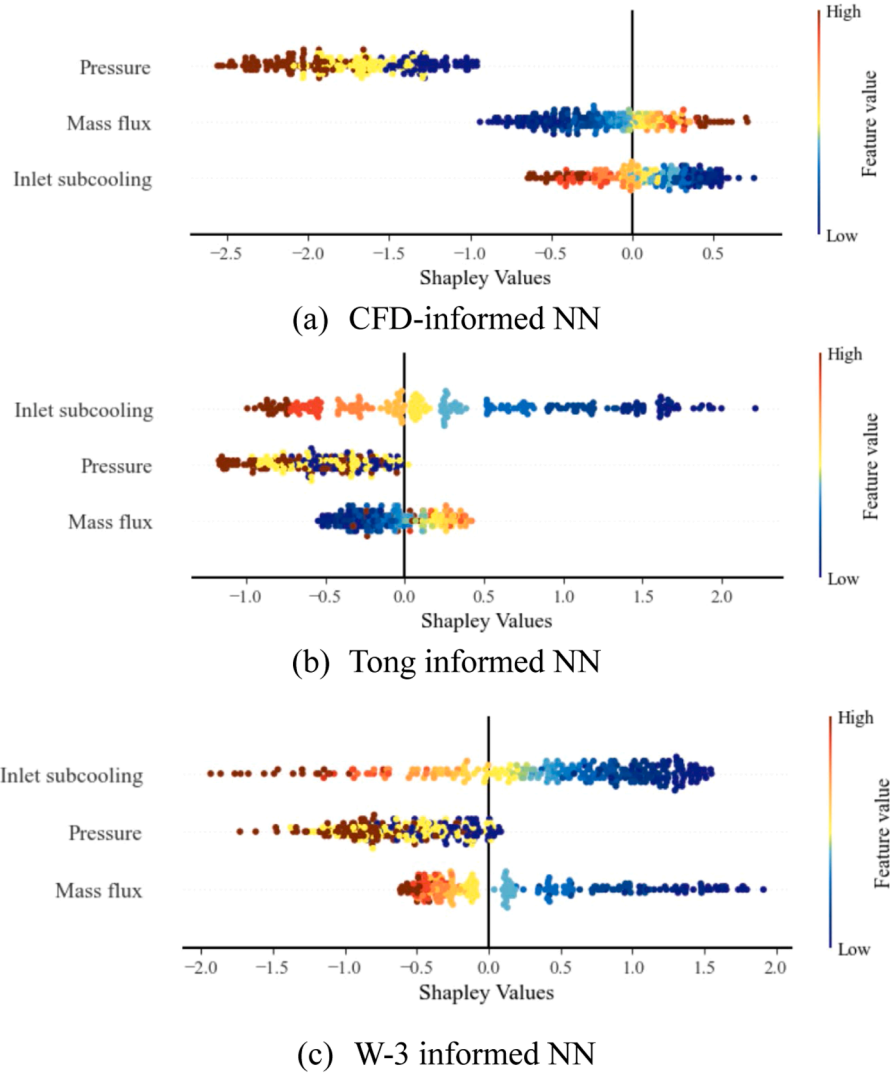
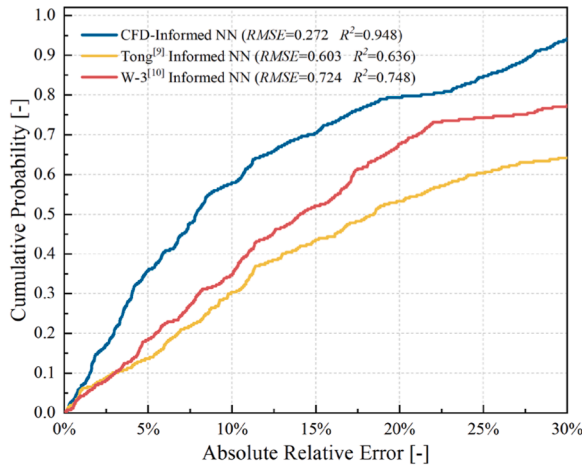


Fig. 22. SHAP feature attribution maps.

Fig. 23. ECDF comparison under $P_{cutoff}=14\text{MPa}$: CFD-informed NN, W-3 informed NN and Tong informed NN.

when contrasted with conventional correlations. Consequently, the CFD-informed NN—specifically designed to account for deviations in CFD predictions—demonstrates superior generalization capabilities on the test dataset.

5. Conclusions

This research employed an advanced heat flux partitioning model integrated within a multiphase CFD program to evaluate the uncertainties inherent in CHF predictions across various criteria. Furthermore, we developed a CFD-informed NN, representing a novel approach that learns from the predictive deviations exhibited by the CFD model. A comprehensive comparative analysis was performed between this novel framework and the PIML method, which leverages the deviations from empirical correlation predictions. The primary conclusions are summarized as follows:

1. Discrepancies and controversies related to relevant bubble dynamics parameters introduce significant uncertainty in CHF predictions employing multiphase CFD methods, underscoring the urgency of delving into the underlying mechanisms that precipitate the boiling crisis.
2. The CFD-informed NN leverages the rich physical information contained within CFD models, exhibiting considerable advantages in

generalization capabilities, especially in extrapolation, compared to standalone neural networks.

3. The robust generalization ability of the CFD-informed NN can be attributed to the sophisticated boiling models within the CFD framework, which provide nonlinear transformations of input features, rendering the CFD prediction deviations more regular compared to the raw CHF data, thereby facilitating the learning process for the neural network.
4. As a repository of domain knowledge, the CFD model encompasses a greater depth of physical information than data-driven empirical correlations, such as flow field and void fraction, resulting in the CFD-informed NN outperforming physics-informed machine learning (PIML) in predictive performance.

The CFD-informed NN integrates critical information pertaining to boiling crises derived from the CFD framework, encompassing analysis of flow fields and void fractions. This integration empowers the CFD-informed NN to extrapolate predictions from low-pressure training datasets to high-pressure test datasets, effectively bridging the gap between data-driven methodologies and the underlying mechanisms that govern CHF. Additionally, the CFD-informed NN can serve as a valuable resource for informing CHF experimental designs while also offering engineering strategies for CHF prediction. Such advancements are imperative for ensuring both the safe operation and economic viability of boiling thermodynamic systems. Future research should prioritize improving the representation of boiling crisis phenomena in CFD models and exploring the predictive capabilities of CFD-informed NN across a wider range of operational conditions.

CRediT authorship contribution statement

Shihao Yang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Xiaoling Li:** Writing – original draft, Methodology, Investigation, Data curation. **Bo Ren:** Software, Methodology, Investigation. **Lixin Yang:** Supervision, Methodology, Investigation. **Chong Chen:** Validation, Software, Funding acquisition. **Qi Lu:** Resources, Project administration, Funding acquisition. **Zonglan Wei:** Visualization, Methodology, Conceptualization.

Declaration of competing interest

The authors declare no known financial conflicts of interest or personal relationships that could have influenced the research presented in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijheatmasstransfer.2025.127429](https://doi.org/10.1016/j.ijheatmasstransfer.2025.127429).

Data availability

Data will be made available on request.

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